

Arthur Cancellieri Pires

Data Scientist / PhD Candidate

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Public project: Industrial Sensor Anomaly Detection API

Professional Experience

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- **Railway Laboratory (Lafer) – partnership with VALE** On-site
LEAD DATA SCIENTIST *January 2022 to Present*
 - Led a team of 12 delivering a predictive maintenance program for VALE, one of the world's largest mining and logistics companies, converting stakeholder needs into shipped ML systems.
 - Cut equipment failure detection latency from 3 months to real time at 91% precision using deep learning models and automated data quality checks on multivariate sensor streams.
 - Flagged critical defects weeks before scheduled maintenance, including cases absent from the maintenance database, with unsupervised Transformer autoencoders on time series data.**Skills:** Python | PyTorch | PyTorch Lightning | scikit-learn | Deep Learning | Transformers | Autoencoders | Anomaly Detection | Time Series | Deployment | FastAPI | Docker
 - **Railway Laboratory (Lafer) – partnership with VALE** On-site
RESEARCH DATA SCIENTIST *June 2020 to January 2022*
 - Built a daily ETL pipeline processing 20,000-40,000 sensor records per asset per day, integrating multiple external sources into one analytical dataset that powered all downstream models.
 - Reduced reliance on costly specialized inspections by predicting asset condition at an R2 of about 0.98 with deep learning models on sensor data.**Skills:** Python | scikit-learn | PyTorch | SQL | PostgreSQL | ETL | Deep Learning | Signal Processing
 - **Tribology and Railway Dynamics Laboratory (LabTDF) – partnership with VALE** On-site
MACHINE LEARNING INTERN *Jan 2019 to Jun 2020*
 - Estimated hard-to-measure contact forces at an R2 of about 0.97 from onboard sensors using a simulation-validated ML pipeline, enabling VALE's predictive maintenance program on instrumented wagons.
 - Optimized wheel/rail wear profiles with genetic algorithms, cutting wear 20% and improving a key safety ratio 35% without sacrificing fatigue performance.
 - Proved the case for a maintenance threshold change through wear/fatigue analysis, driving adoption into operations and extending service life 29%-50% (29% from the threshold change alone, up to 50% when combined with the optimized wheel profile)**Skills:** Python | MATLAB | Machine Learning | scikit-learn | Optuna | Genetic Algorithms | Optimization | Multibody Simulation

Education

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- **University of Campinas (UNICAMP)** Campinas, Brazil
PH.D. IN MECHANICAL ENGINEERING *Expected: August 2026*
 - **Thesis:** A methodology for detecting track defects from instrumented railway vehicle data.
 - **University of Campinas (UNICAMP)** Campinas, Brazil
M.SC. IN MECHANICAL ENGINEERING *Jan 2022*
 - **Thesis:** Measuring railway track irregularities from instrumented railway vehicle data using machine learning techniques.
 - **Federal University of Espírito Santo (UFES)** Vitória, Brazil
B.SC. IN MECHANICAL ENGINEERING *May 2020*
 - **Thesis:** Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning.

Languages

Portuguese (Native), English (Fluent), Italian (Basic)

Publications & Conferences

SELECTED PUBLICATIONS

- 2024 **Measuring vertical track irregularities from instrumented heavy haul railway vehicle data using machine learning**, Engineering Applications of Artificial Intelligence
- 2023 **Influence of wheel tread wear on Rolling Contact Fatigue and on the dynamics of railway vehicles**, Wear
- 2022 **Nonintrusive Operator Inference for Chaotic Systems**, IEEE Transactions on Artificial Intelligence
- 2022 **Generation of realistic artificial track irregularities for multibody simulation using measured geometric data: from mid-chord to space-curve**, In Proceedings of the Fifth International Conference on Railway Technology
- 2021 **Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning**, Mechanical Systems and Signal Processing
- 2021 **The effect of railway wheel wear on reprofiling and service life**, Wear

CONFERENCE PRESENTATIONS

- 2026 International Conference on Railway Technology (Railways 2026), Aug. 23–26 (accepted) *Budapest, Hungary*
- 2025 International Heavy Haul Association (IHHA) Conference, Nov. 17–21 *Colorado Springs, USA*
- 2023 International Heavy Haul Association (IHHA) Conference, Aug. 27–31 *Rio de Janeiro, Brazil*